

Code App with Dataverse

Power CAT | The Intelligent Enterprise - Power Platform & AI for Frontier Firms

Code App with Dataverse

Intelligent Apps and Agents — Lab authoring template

Level	2 0 0
Persona	Pro Code / Maker
Estimated duration	4 5 minutes
Audience assumptions	Familiarity with Dataverse, Code Tools, and basic web app development
Author / team	Christopher Moncayo
Last updated	2 0 2 6 - 0 1 - 2 8
Version	[v 0 . 1]

1 . Lab overview

1 . 1 Lab purpose

This lab demonstrates the **Bring Your Own Code (BYOC)** feature in **Power Apps**, enabling developers to integrate custom logic into a single page application for **Supplier Onboarding Management**. The backend system leverages **Microsoft Dataverse** and connects to the **Suppliers** table to retrieve and manage supplier records.

The app displays supplier data in a tabular format, allowing users to:

- **View Details:** Click on a supplier record to open a **modal dialog** showing detailed information.
- **Take Action:** Approve or decline onboarding requests directly from the modal interface. This scenario highlights how BYOC extends Power Apps functionality by combining custom code with Dataverse data, delivering a tailored experience for supplier management workflows.

1 . 2 Why this matters

The BYOC feature empowers development teams to **reuse existing code** or **create bespoke experiences** with custom logic, reducing duplication and accelerating delivery. It also enables teams to integrate their own code seamlessly with tools they already know and trust—such as **Visual Studio Code**, **Git**, and **GitHub Copilot**—bringing modern development practices into the Power Platform ecosystem. This flexibility fosters innovation, improves maintainability, and ensures alignment with enterprise coding standards.

2 . App scenario

Industry / function	Operations
Primary user role	Operations Manager
Problem statement	How can Contoso Electronics use their existing code in Power Apps
Intelligent outcome	Bring your own code allows the enterprise to run their own custom code within the Power Apps environment

3 . Core concepts

Concept	Why it matters
Bring Your Own Code	Publish bespoke code to Power Apps
Governance & ALM	Enterprise can use their own ALM against their bespoke code.

4 . Intelligence used in this lab

Intelligence type	Where it's used	Purpose
Github Copilot	Visual Studio Code	Provide a code agent to help the developer quickly code their application

5 . Documentation & learning resources

6 . Prerequisites

Import Northwind Traders version > 0 . 1 1 to the environment and seed with data.

Long term support (LTS) version

Power Platform CLI

End users that run code apps need a Power Apps Premium License

7 . Learning outcomes

You're able to create a web app using VS Code, Power Platform SDK.

You're able to connect your created web application to a Dataverse table.

You're able to update your app using Github Copilot to facilitate Supplier Onboarding.

You're able to publish your app to Power Platform and view it in the browser.

8 . Use cases covered

Use case	Description	Value	Est. time
Create web app	Clone the app from the templates and publish to Power Apps.	Scaffolding needed to get the app started.	1 5 min
Add data source	Add the suppliers table to the app.	Connect to the Dataverse back end.	5 min
Update app to include business logic	Use Github Copilot to update the app for the Suppliers onboarding business logic.	Apply the business logic to cover business logic and user stories.	2 0 min

9 . Lab instructions

Use case 1 : Create a code app from scratch

Use case value	Scaffolding to get your app started
Estimated effort	5 minutes
Scenario	Would be required to connect your app to the Dataverse environment.
Objective	Clone the repo, connect the app to Dataverse,

Tasks covered

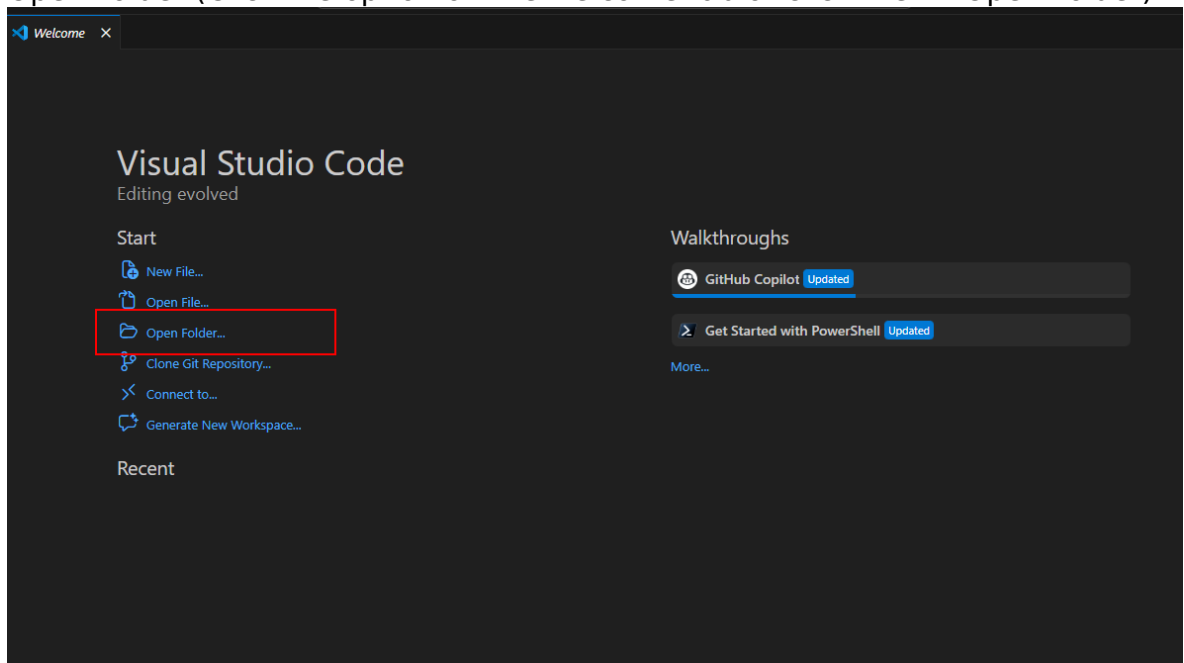
Clone the starter template into a new folder.

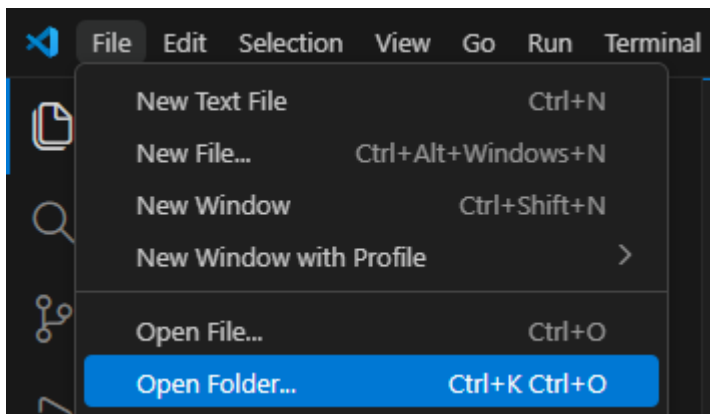
Connect to the Dataverse Environment.

Build and deploy the app to Power Apps within a solution

Step-by-step instructions

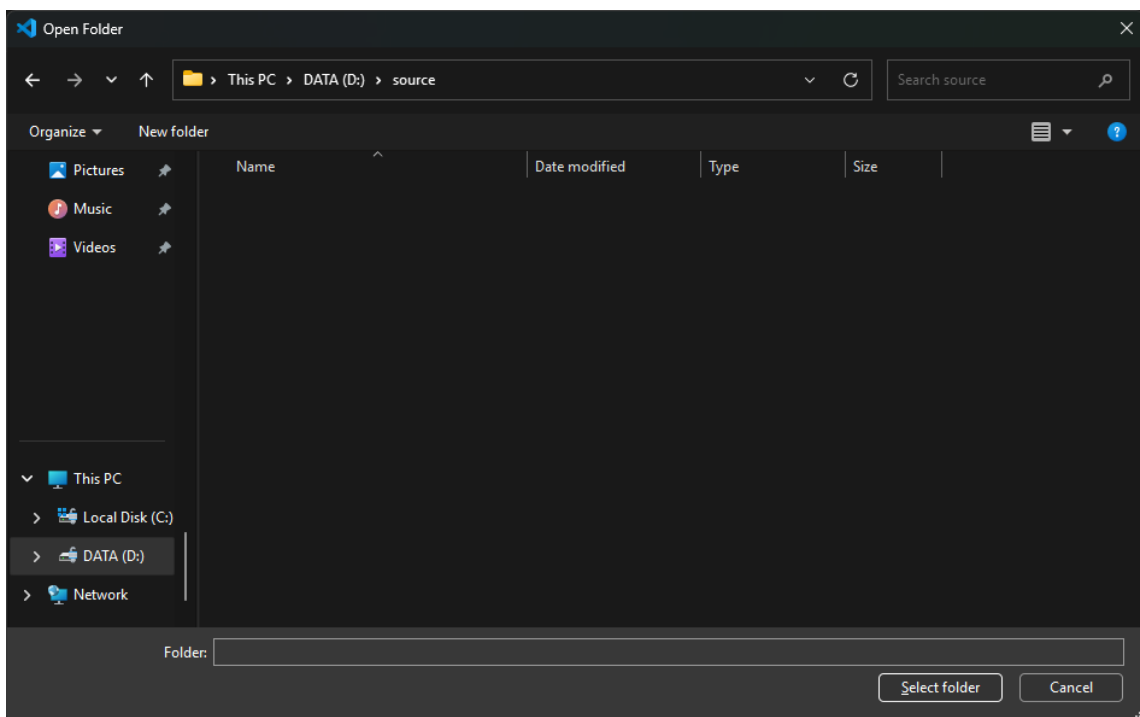
- Open Visual Studio Code.
- Open Folder (Click the option on the welcome tab or click File -> Open Folder)





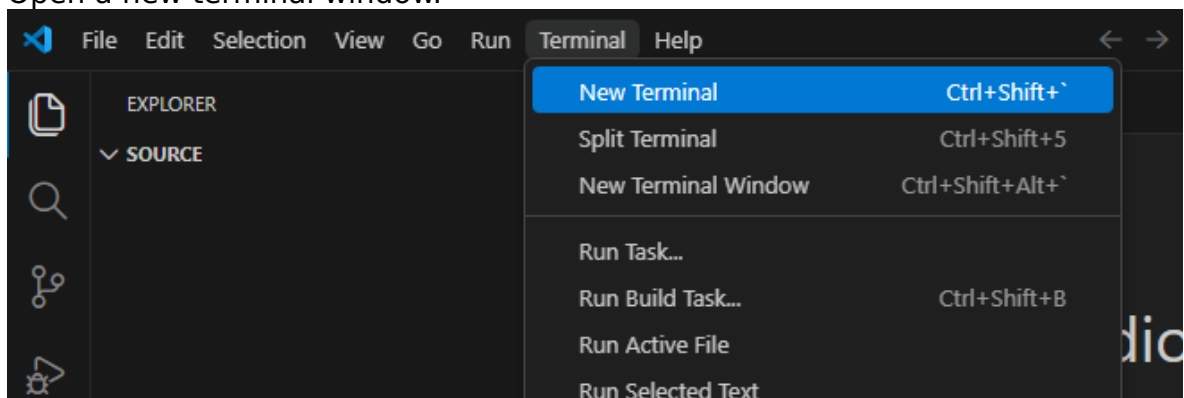
open folder click the option on the welcome tab or click file open fol 0 2

And open your folder where you want to host the source code.



and open your folder where you want to host the source code 0 3

- Open a new terminal window.



- Let's get a copy of a starting point for a Code App from the Microsoft Sample repository.

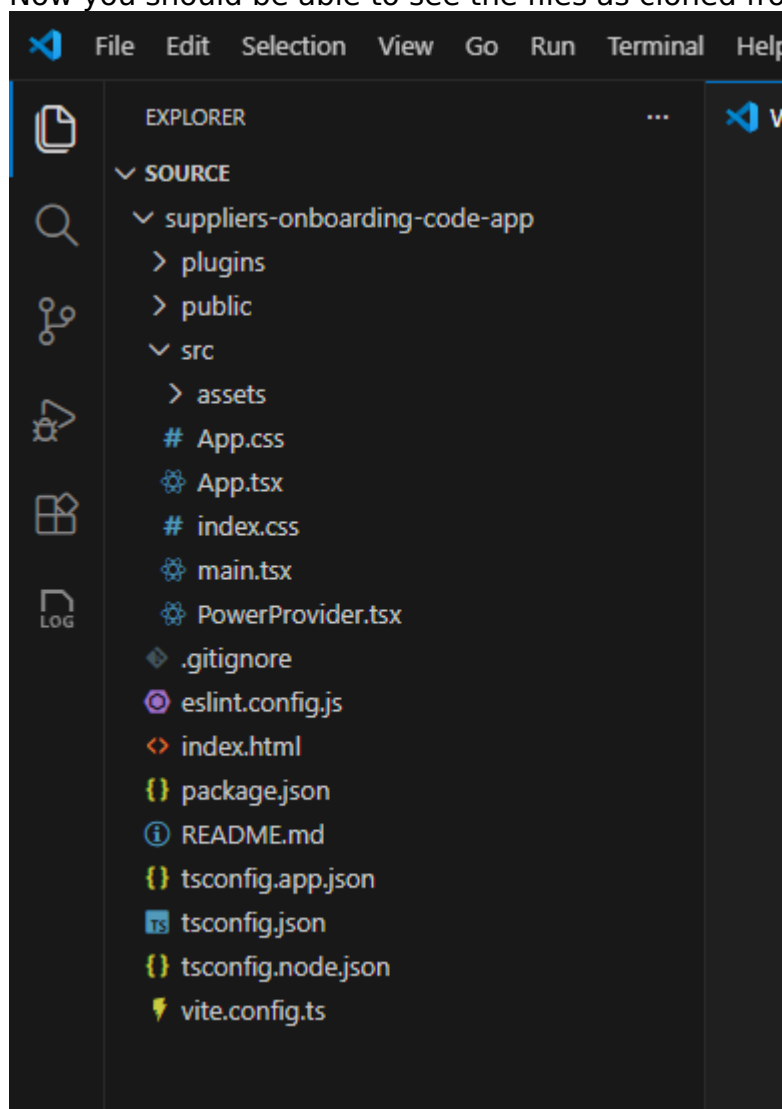
```
npx degit github:microsoft/PowerAppsCodeApps/templates/vite suppliers-  
onboarding-code-app  
cd suppliers-onboarding-code-app
```

```
PS D:\source> npx degit github:microsoft/PowerAppsCodeApps/templates/vite suppliers-onboarding-code-app  
(node:16716) Warning: `--localstorage-file` was provided without a valid path  
(Use `node --trace-warnings ...` to show where the warning was created)  
> cloned microsoft/PowerAppsCodeApps#HEAD to suppliers-onboarding-code-app  
PS D:\source> cd suppliers-onboarding-code-app
```

lets get a copy of a starting point for a code app from the microsoft 0 5

 **Tip:** Run these commands from your local source folder. Instead of “suppliers-onboarding-code-app” you can also specify a name you want to use.

- Now you should be able to see the files as cloned from the repo.



Source code will be in the src folder, these contain all the files for your application.

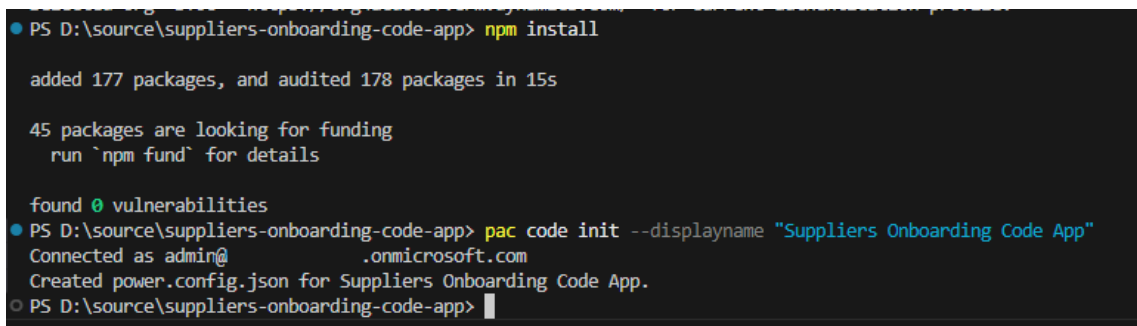
- Authenticate the Power Platform CLI against your Power Platform tenant and select your environment:

```
pac auth create
pac env select --environment < your-environment-ID >
```

 **Important:** Sign in using your Power Platform account when prompted. The PAC CLI's auth command prompts you to authenticate using your Microsoft Entra identity. The environment specified in "your-environment-id" ensures that the connections added to your code app and publish to Power Platform go to the specified environment.

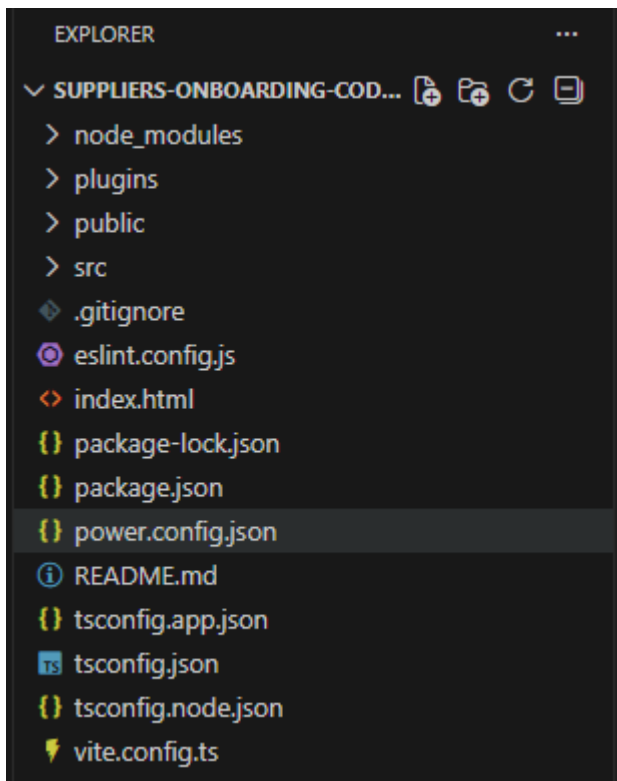
- Install the Power SDK and initialize your code app by using:

```
npm install
pac code init --displayname "Suppliers Onboarding Code App"
```



```
PS D:\source\suppliers-onboarding-code-app> npm install
added 177 packages, and audited 178 packages in 15s
45 packages are looking for funding
  run `npm fund` for details
found 0 vulnerabilities
PS D:\source\suppliers-onboarding-code-app> pac code init --displayname "Suppliers Onboarding Code App"
Connected as admin@.onmicrosoft.com
Created power.config.json for Suppliers Onboarding Code App.
PS D:\source\suppliers-onboarding-code-app>
```

install the power sdk and initialize your code app by using 0 7



```
EXPLORER
SUPPLIERS-ONBOARDING-COD...
  > node_modules
  > plugins
  > public
  > src
  .gitignore
  eslint.config.js
  index.html
  package-lock.json
  package.json
  power.config.json
  README.md
  tsconfig.app.json
  tsconfig.json
  tsconfig.node.json
  vite.config.ts
```

install the power sdk and initialize your code app by using 0 8



Important: power.config.json contains all the settings for connecting to the Power Platform including appDisplayName, environmentId, connectionReferences and databaseReferences.

```
{ } power.config.json X
{ } power.config.json > ...
1  {
2    "appId": "",
3    "appDisplayName": "Suppliers Onboarding Code App",
4    "description": null,
5    "environmentId": "5d19f100-8b76-ee23-9141-11d795e92810",
6    "buildPath": "./dist",
7    "buildEntryPoint": "index.html",
8    "logoPath": "Default",
9    "localAppUrl": "http://localhost:3000/",
10   "connectionReferences": {},
11   "databaseReferences": {}
12 }
```

install the power sdk and initialize your code app by using 0 9

- Enter the following command to test your code app locally:

```
npm run dev
```

- Then, open the URL labelled **Local Play**.

```
PS D:\Src\CodeApp-SupplierOnboardingBackend> npm run dev

> vite@0.0.0 dev
> vite

Power Apps Vite Plugin

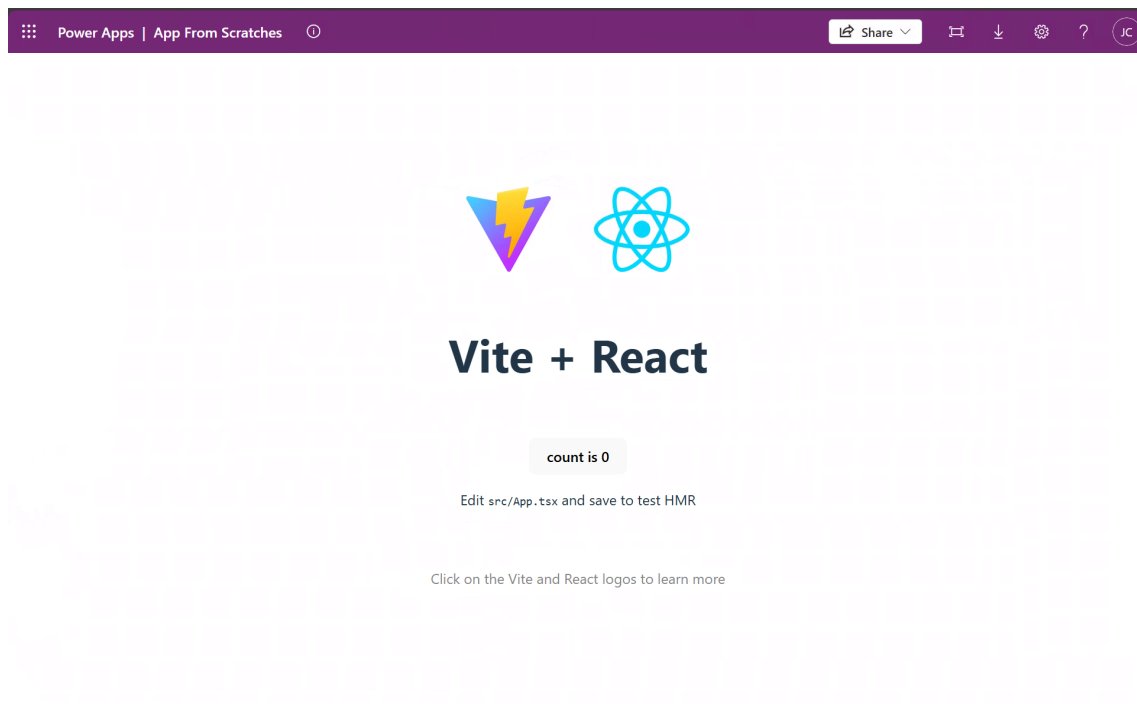
→ Local Play: https://apps.powerapps.com/play/e/2f458b60-a296-e9b2-9e3a-16d298b5fb0d/a/local?_localAppUrl=http://localhost:5173/&_localConnectionUrl=http://localhost:5173/_vite_powerapps_plugin_/power.config.json

VITE v7.3.1 ready in 2401 ms

→ Local: http://localhost:5173/
→ Network: use --host to expose
→ press h + enter to show help
TIP: [Add best practices, common mistakes, and prompt guidance]
```

Expected result (checkpoint)

You should see the app open like:



you should see the app open like 1 1

 **Important:** This is the default vite template, we will be updating this to have our business logic in the third module.

 **Tip:** To exit the local webserver, press CTRL+C in the terminal window.

- Build and deploy the app to Power Apps. In the terminal window, run these commands:

```
npm run build | pac code push --solutionName NorthwindTraders
```

- Runs the scripts configured in the package.json file with the key value of build. In this case, the script is "tsc -b && vite build".
- Publishes a new version of a code app.
- targets a specific solution, in this case, we'll push the code app back to Northwind Traders solution. >  **Important:** If there is no solution specified, pac code push will deploy the app in the default or preferred solution in the environment. Also, if you've previously deployed without using the solutionName flag, you will need to delete your app from Power Apps, delete the power.config.json file and reinitialize it.

If successful, this command returns a Power Apps URL to run the app.

```
PS D:\source\suppliers-onboarding-code-app> npm run build | pac code push --solutionName NorthwindTraders
Connected as admin@B365x49237538.onmicrosoft.com
Saving app ...
Connected to... BYOC

Listing all Solutions from the current Dataverse organization...
bolt.system.VerBExecutionException: CodeSolutionNotFound/Message
    at bolt.module.code.PushVerb.EstablishInputSolution(String inputSolutionName, SelectedOrganizationInfo selectedOrganizationInfo) in E:\PowerPlatform-
\bolt.module.code\verbs\PushVerb.cs:line 172
    at bolt.module.code.PushVerb.ExecuteAsync(Command command, CancellationToken cancellationToken) in E:\PowerPlatform-Scale-AdminTools\src\cli\bolt.mod
cs:line 108 Failed to push the app. Retrying...
App has successfully been pushed!
You can test the app at the following url https://apps.powerapps.com/play/e/5d19f100-8b76-ee23-9141-11d795e92810/a/0d5bc453-1b0a-4859-800f-74be7c60d4ff
PS D:\source\suppliers-onboarding-code-app>
```

if successful this command returns a power apps url to run the app 1 2

Optionally, you can open Power Apps to see the app. You can play, share, or see details from there.

Reflection

You've completed the first part of the lab, you successfully pushed your code app.

Use case 2 : Add Dataverse Table to the App

Use case value	Adding data sources to make your app more robust.
Estimated effort	5 minutes
Scenario	Adding the Suppliers table to the app
Objective	Learn how to add Dataverse tables to a code app.

Tasks covered

Use Power Shell to add your data source

Push the code back to Power Apps

Step-by-step instructions

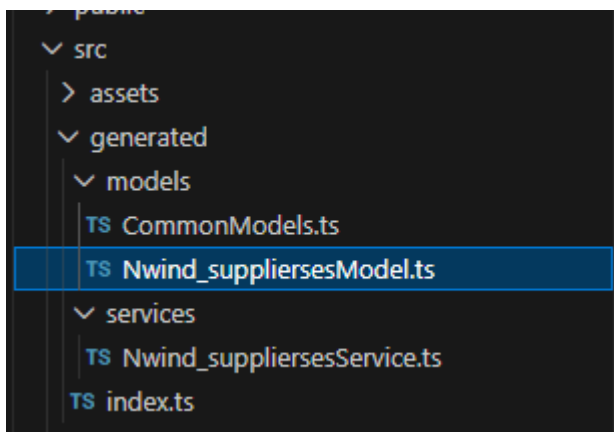
- 1 . Ensure you're connected to your environment using PAC CLI. If you're not connected, see **Use Case 1 - Step 2** for instructions.
- 2 . Within a terminal, use the **pac code add-data-source** command to add Dataverse as a data source to your code app.

```
pac code add-data-source -a dataverse -t nwind_suppliers
```



Important: The syntax for the table is to use the logical name for the Dataverse table you want to connect to.

You will know this has completed successfully when you can see generated code now appear in your folder.

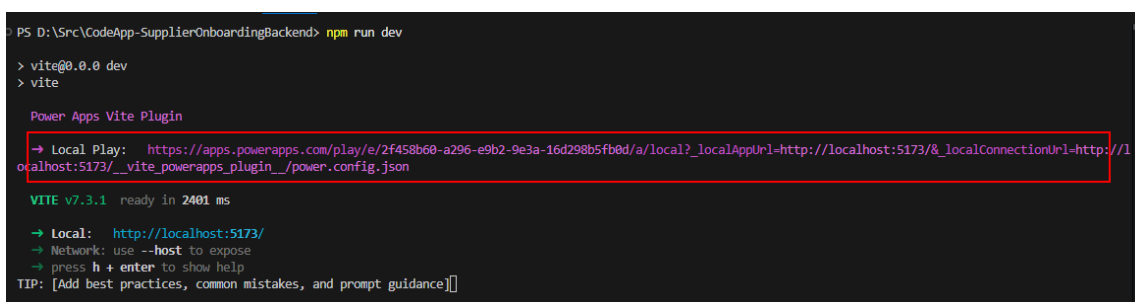


you will know this has completed successfully when you can see generat
1 3

- 1 . Once the table has been added to your application, enter the following command to test your code app locally:

```
npm run dev
```

Copy and paste the “Local Play” link from the terminal into your browser.



then open the url labelled local play 1 0

Expected result (checkpoint)

Your app is still working and now has the Dataverse Suppliers table connected to it.

Reflection

We’re now ready to start adding business logic to the app.

Use case 3 : Use Github Copilot to add business logic

Use case value	Add business logic to the starter template
Estimated effort	2 0 minutes
Scenario	Adding business logic to the app to reflect the user stories that would satisfy the objective for the app.

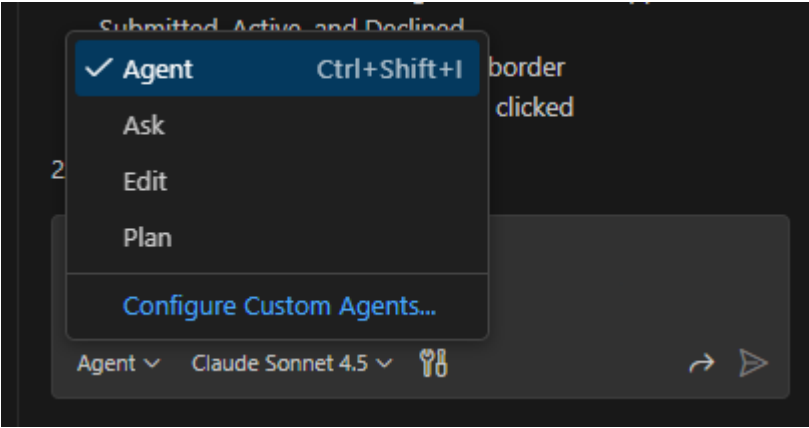
Use case value	Add business logic to the starter template
Objective	Use Github Copilot Code Agent to update the app and add business logic to the application.

Tasks covered

- Use Github Copilot to add business logic.
- Test the application to ensure it works.
- Build and deploy the app to Power Apps.

Step-by-step instructions

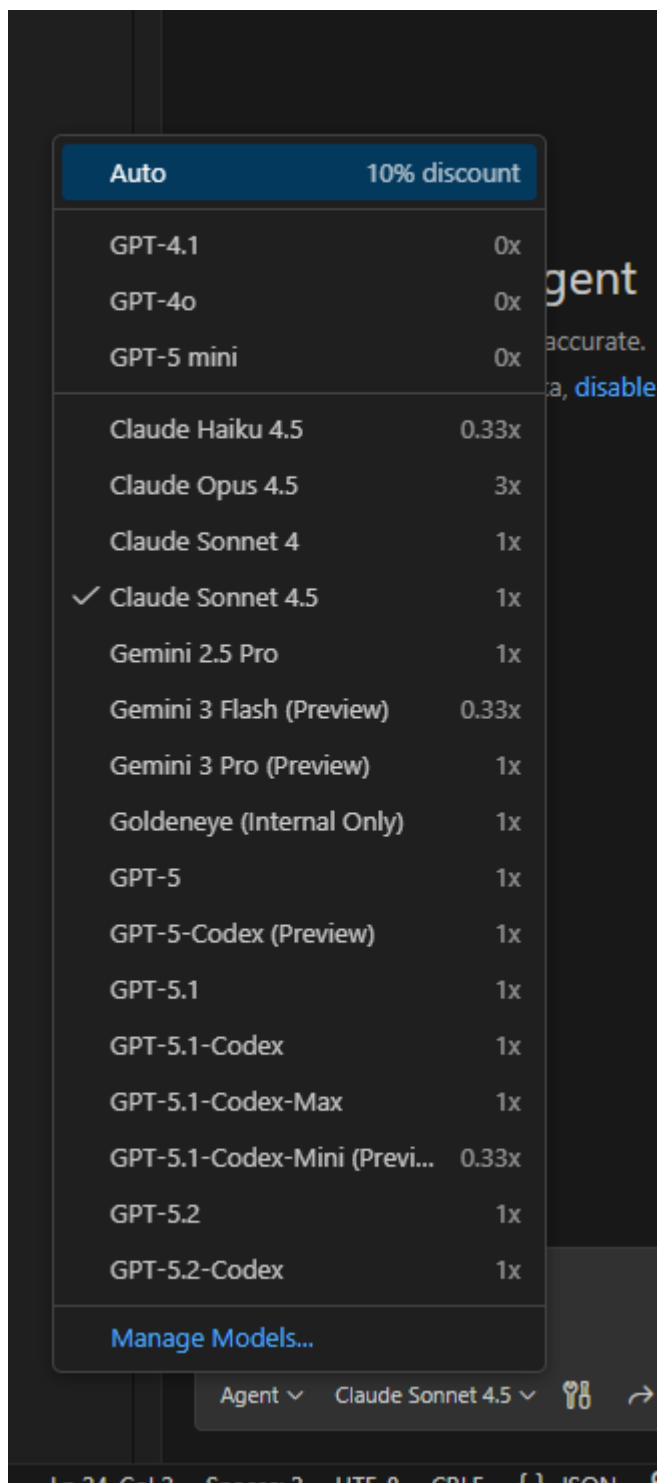
- Ensure Github Copilot chat is in Agent Mode and select the model you want to use.



ensure github copilot chat is in agent mode and select the model you w

1 4

Github Copilot also provides several models you can use with your agent. In our lab, we're using Claude Sonnet 4 . 5



github copilot also provides several models you can use with your agent

- Execute the following prompt:

Update the application to be a backend supplier onboarding acceptance tool. List all the suppliers in a grid. Make the page look like a dashboard which has card buttons which will filter the list for the different status reasons for the suppliers: Submitted, Active, Declined. The items in the grid should be clickable and on click, they should show in a modal dialog. The modal dialog should show the full details of the supplier, and an edit button to let them edit any values and save them back to the database. There should also be an accept and decline button to let the user accept or decline from the modal

dialog. Ensure the app is responsive for mobile devices. Optimize for iPhone 12.



Tip: Depending on which model you use, you may get different reasoning and steps the agent will take to update the app.

Here is an example of the chat results:

```
> Used 1 reference
✓ Read App.tsx and analyzed data structure

I'll help you build this supplier onboarding backend tool. Let me
first examine the existing code structure and then implement
the dashboard with filtering, grid, and modal dialog
functionality.

✓ Read application and model service files

Now let me create a comprehensive supplier onboarding
dashboard. I'll update the App.tsx and App.css files to
implement all the requested features.

✓ Edited App.tsx +454 -24

Now let me update the CSS file to style the dashboard, grid, and
modal dialog:

✓ Edited App.css +384 -35

Perfect! I've successfully transformed your application into a
comprehensive Supplier Onboarding Dashboard. Here's what
has been implemented:

✓ Features Implemented

1. Dashboard with Filter Cards

• Four clickable cards showing counts for: All Suppliers,
  Submitted, Active, and Declined
• Active card highlights with a blue border
• Cards filter the supplier grid when clicked

2. Supplier Grid

• Displays all suppliers in a clean, professional table
```

here is an example of the chat results 1 6



Important: Your app may have errors as marked in red. You can ask the agent to fix the errors, by simply saying “fix the errors” or prompting the agent to “run lint” which will do its best to do linting against the solution.

- Once the agent is finished coding the app and you’re free of errors, you can test the app locally by running the following command in the terminal:

```
npm run dev
```



Tip: If npm run dev doesn’t show you any data, or you are getting javascript errors in the app, you can copy and paste the errors from the console and put them in the Github Copilot Chat box and it will try and fix the errors.

Copy and paste the “Local Play” link from the terminal into your browser.

```
PS D:\Src\CodeApp-SupplierOnboardingBackend> npm run dev
> vite@0.0.0 dev
> vite

Power Apps Vite Plugin

→ Local Play: https://apps.powerapps.com/play/e/2f458b60-a296-e9b2-9e3a-16d298b5fb0d/a/local?_localAppUrl=http://localhost:5173/&_localConnectionUrl=http://localhost:5173/_vite_powerapps_plugin_/power.config.json

VITE v7.3.1 ready in 2401 ms
→ Local: http://localhost:5173/
→ Network: use --host to expose
→ press h + enter to show help
TIP: [Add best practices, common mistakes, and prompt guidance]
```

then open the url labelled local play 1 0

The screenshot shows a web browser displaying the 'Supplier Onboarding Dashboard' for 'Contoso Electronics'. The dashboard has a purple header bar with the title 'Supplier Onboarding Dashboard'. Below the header, there are four summary cards: 'All Suppliers' (10), 'Submitted' (9), 'Active' (1), and 'Declined' (0). Below these cards is a table with columns: COMPANY, CONTACT NAME, EMAIL, PHONE, CITY, and STATUS. The table lists eight suppliers, with their status indicated by a colored pill (green for Active, yellow for Submitted).

COMPANY	CONTACT NAME	EMAIL	PHONE	CITY	STATUS
Supplier J	Luis Sousa	-	-	-	ACTIVE
Supplier I	Mikael Sandberg	-	-	-	SUBMITTED
Supplier H	Bryn Paul Dunton	-	-	-	SUBMITTED
Supplier G	Stuart Glasson	-	-	-	SUBMITTED
Supplier F	Satomi Hayakawa	-	-	-	SUBMITTED
Supplier E	Amaya Hernandez-Echevarria	-	-	-	SUBMITTED
Supplier D	Naoki Sato	-	-	-	SUBMITTED
Supplier C	Madeleine Kelley	-	-	-	SUBMITTED

copy and paste the local play link from the terminal into your browser 1 7



Important: Your app may have errors shown on the screen or in the JavaScript console. Copy any errors from the console or browser and ask the agent to fix the errors, by simply saying “fix this error” and paste the error message or prompting the agent to “run lint” which will do its best to do linting against the application.

- Keep iterating with the code agent until your app is satisfactory for you.
- Once you’re satisfied with the app, build and deploy your app to Power Apps using the following command:

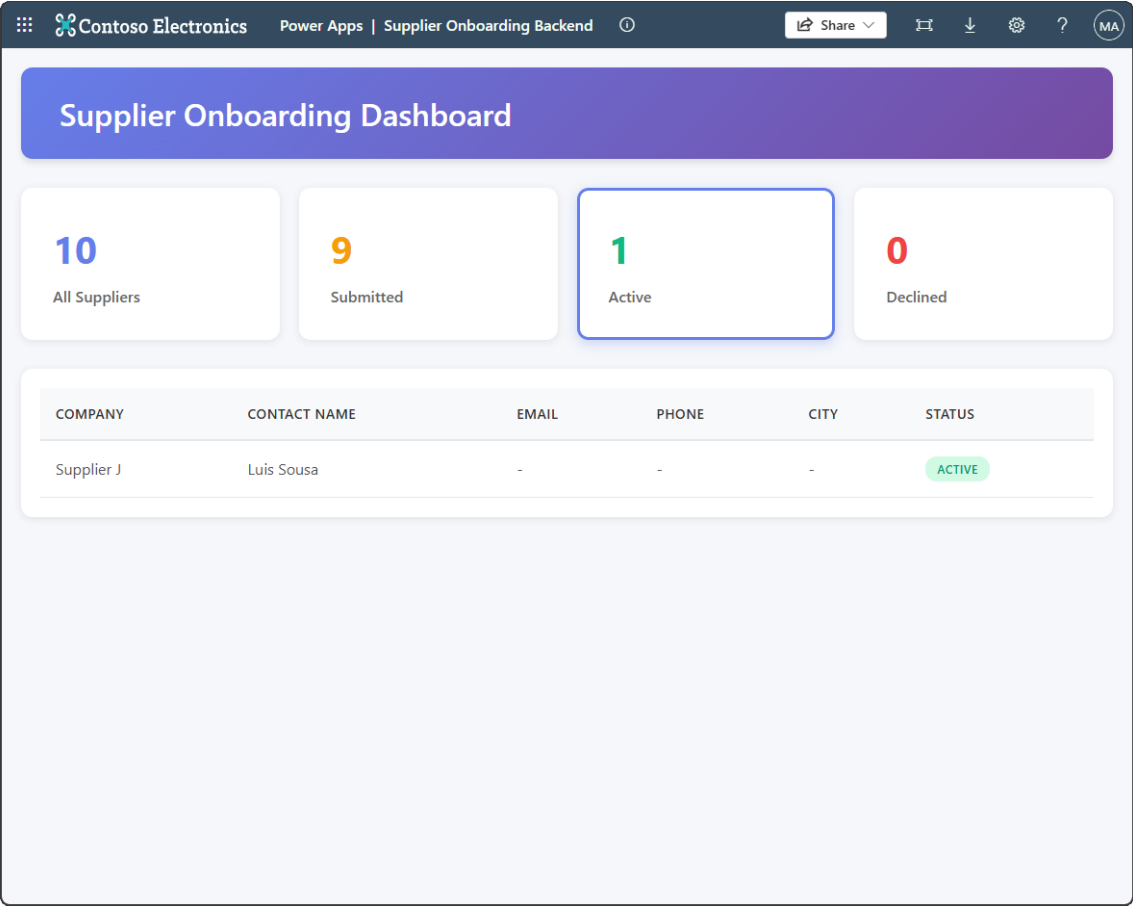
```
npm run build | pac code push
```

Expected result (checkpoint)

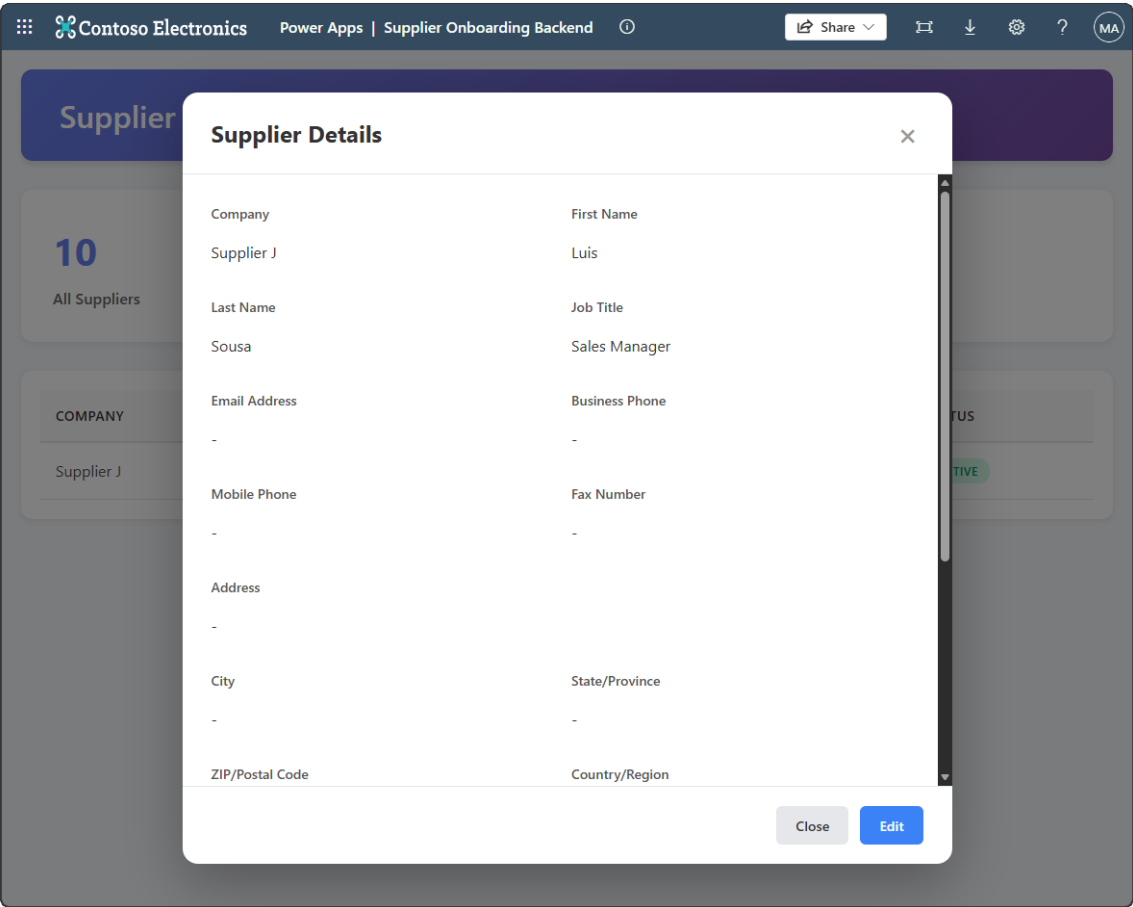
At this point your application should now be able to data entry and retrieval against the Suppliers table in a dashboard style application.

COMPANY	CONTACT NAME	EMAIL	PHONE	CITY	STATUS
Supplier J	Luis Sousa	-	-	-	ACTIVE
Supplier I	Mikael Sandberg	-	-	-	SUBMITTED
Supplier H	Bryn Paul Dunton	-	-	-	SUBMITTED
Supplier G	Stuart Glasson	-	-	-	SUBMITTED
Supplier F	Satomi Hayakawa	-	-	-	SUBMITTED
Supplier E	Amaya Hernandez-Echevarria	-	-	-	SUBMITTED
Supplier D	Naoki Sato	-	-	-	SUBMITTED
Supplier C	Madeleine Kelley	-	-	-	SUBMITTED

copy and paste the local play link from the terminal into your browser 1 7



at this point your application should now be able to data entry and re 1 8



at this point your application should now be able to data entry and re 1 9

Contoso Electronics Power Apps | Supplier Onboarding Backend

Share

Supplier Details

Company	First Name
Supplier I	Mikael
Last Name	Job Title
Sandberg	Sales Manager
Email Address	Business Phone
-	-
Mobile Phone	Fax Number
-	-
Address	
-	
City	State/Province
-	-
ZIP/Postal Code	Country/Region

Close Edit Decline Accept

at this point your application should now be able to data entry and re 2 0

1 1 . Lab completion

Congratulations! You've built a Code App connected to a Dataverse Table.

Key takeaways

You can now create an app, connect it to Dataverse and publish it to Power Apps.

Power Apps code apps let developers bring Power Apps capabilities into custom web apps built in a code-first IDE.

Develop locally and run the same app in Power Platform.

Build with popular frameworks while keeping full control over your UI and logic.

1 2 . Challenge: apply this to your scenario

What data would improve AI accuracy?

Where could Copilot reduce user effort?

What governance controls are critical?

1 3 . Summary & best practices

Intelligent App golden rules:

Design AI-first, not AI-last

Ground every response

Combine deterministic + generative intentionally

Test, monitor, and iterate

Design for trust

Appendix A — 1 0 0 -level maker quick guidance

Use this appendix to keep the lab friendly for makers and first-time Intelligent Apps builders.



Tip: Keep each step under one screenful. Prefer guided configuration over deep theory. Use screenshots for every “where do I click?” moment.

A. 1 Maker prompts and wording

Use plain language

Avoid unnecessary jargon

Make checkpoints obvious (what should I see?)

A. 2 Common troubleshooting

Wrong environment selected

Missing license or permissions

DLP blocks connector

Sample data missing or not loaded